- Then raise the tube approximately about two times the first resonating length. Excite the tuning fork again and place it on the open end of the tube.
- Adjust the height of the air column until the maximum sound is heard.
- Measure the length of air column at this position. This is taken as the second resonating length l_2
- We can now calculate the velocity of sound in air at room temperature by using the relation.

  $V = 2\nu(l_2 - l_1)$
- Repeat the experiment with tuning forks of different frequency and tabulate the corresponding values of l_1 and l_2.
- The mean of the calculated values will give the velocity of sound in air at room temperature.

OBSERVATIONS

Sl. No.	Frequency of tuning fork 'ν' (Hz)	First resonating length l_1 $(\times 10^{-2} m)$			Second resonating length l_2 $(\times 10^{-2} m)$			$l_2 - l_1$ $(\times 10^{-2} m)$	Velocity of sound $V = 2\nu(l_2 - l_1)$ (m s^{-1})
		Trial 1	Trial 2	Mean	Trial 1	Trial 2	Mean		
1									
2									
3									
								Mean V =	

CALCULATION

Velocity sound in air at room temperature, $V = 2\nu(l_2 - l_1)$ = _____ m s^{-1}

RESULT

Velocity of sound in air at room temperature, is found to be (V) = _____ m s^{-1}

UNIT-12(XI-Physics_Vol-2) PRACTICAL FIRST YEAR.indd 294 17-08-2020 16:26:41